

Monepha Mitchell

Professor Viswanatha Rao (Vishwa)

ITAI 1371 Introduction to Machine Learning

27 Jun 2026

Reflective Journal

This lab helped me better understand why data preparation is such an important step before building a machine learning model. I learned that even if a dataset contains useful information, it may have missing values, categorical data, or features with different scales that need to be processed first. Completing each step helped me understand how these preprocessing techniques improve data quality and prepare it for machine learning.

One of the first things I worked on was filling in the missing values in the Age column using the median. I learned that the median is often a better choice than the mean because it is less affected by extreme values. After that, I used One-Hot Encoding to convert the categorical columns into numerical values that a machine learning model can understand. Finally, I applied feature scaling to the Age and Fare columns using StandardScaler. This showed me how scaling helps put numerical features on a similar range, which is important for many machine learning algorithms.

I also enjoyed working with my group during this lab. Although we each completed our own notebooks, we communicated throughout the assignment to discuss any challenges we encountered and to help each other understand the preprocessing steps. Sharing ideas made it easier to complete the lab and ensured that everyone understood the concepts before submitting their work.

Overall, this lab gave me more confidence in using Python to prepare data for machine learning. It reinforced that data cleaning and preprocessing are essential parts of the machine

learning process. I know these skills will be useful in future labs and in our upcoming midterm and final projects, where correctly preparing a dataset will be just as important as building the machine learning model itself.